Efficient Robotic Pushing with Diffusion Policy: Adaptive Execution and Self-Imitation-Based Fine-Tuning

Qihang Feng ECE 730 Project Proposal

Motivation and question. Receding-horizon control executes only part of its predictions, leaving potentially useful movements unused. Stochastic policies can succeed or fail from the same situation, yielding successful trajectories for learning. This project explores how to use predictions more effectively through adaptive execution and self-imitation-based fine-tuning. I will use PushT, a two-dimensional simulator where a pusher moves a T-shaped block onto a target [2]. Camera feedback links physical motion and software decisions in a cyber-physical loop.

Proposed method. I will build on a public pretrained Diffusion Policy [1] with two approaches:

1. **Adaptive execution.** A lightweight controller will assess camera-estimated alignment and remaining predictions, then replan after eight movements or execute up to four more. Confirmed deterioration may trigger bounded early interruption; temporary declines alone will not. Prediction length and inference settings will remain fixed.
2. **Self-imitation-based fine-tuning [3].** From selected intermediate states in failed baseline attempts, I will sample multiple continuations using the original model. Successful recovery trajectories will be mixed with expert demonstrations to fine-tune only the final convolution block, keeping other parameters frozen. The updated model will use the same adaptive controller.

Evaluation plan. I will compare four configurations.

Configuration	Model	Execution strategy
Original baseline	Original pretrained model	Fixed eight movements
Adaptive execution	Original pretrained model	Adaptive movements
Fine-tuning	Fine-tuned model	Fixed eight movements
Adaptive execution + fine-tuning	Fine-tuned model	Adaptive movements

The four metrics will be: success rate, average final alignment, average calls, and average runtime.

Expected contribution. This project aims to improve pushing success while reducing average prediction calls and runtime through adaptive execution and self-imitation-based fine-tuning. Its effect on average final alignment will also be evaluated.

[1] Chi et al. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. RSS, 2023.

[2] Hugging Face. gym-pusht simulator.

[3] Jin et al. SIME: Enhancing Policy Self-Improvement with Modal-level Exploration. IROS, 2025.